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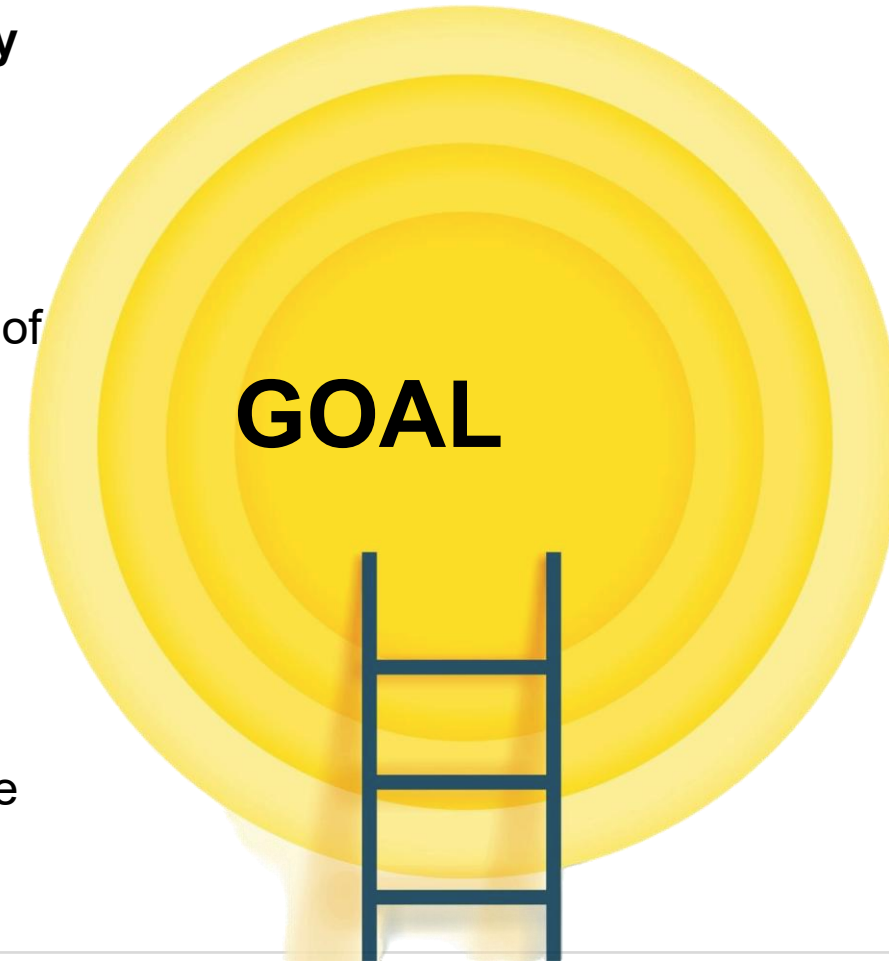
AI-Driven Solar Power Generation Forecasting by- **PULKIT AGRAWAL**

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Learning Objectives

- **Data Preparation Mastery (Week 1):** Successfully executed **Exploratory Data Analysis (EDA)**, handle feature selection (removing low-variance features like precipitation), and establish data integrity.
- **Baseline Establishment (Week 2):** Implement and validate a robust **Random Forest Regressor**, identifying the **most critical features** (Angle of Incidence, Zenith, Shortwave Radiation).
- **Advanced Model Implementation (Week 3):** Master the specialized preprocessing steps (**Data Scaling and Sequence Framing**) required to successfully build and train a **Long Short-Term Memory (LSTM)** neural network.
- **Final Evaluation:** Rigorously compared the final LSTM RMSE against the baseline RMSE to quantify the project's technical success.



Tools and Technology used

- Programming & Core Libraries (Python):

1.) **Pandas & NumPy**: Essential for data loading, cleaning, transformation (Week 1), and converting data into the required 3D arrays for the LSTM model (Week 3).

2.) **Scikit-learn (Sklearn)**: Used for data splitting, implementing the **Random Forest Regressor** (Week 2 Baseline), and calculating performance metrics (RMSE).

- Deep Learning Framework:

1.) **TensorFlow / Keras**: The core framework used to build, compile, and train the advanced **Long Short-Term Memory (LSTM)** Neural Network (Week 3).

- Visualization:

1.) **Matplotlib & Seaborn**: Used for Exploratory Data Analysis (EDA) and plotting the crucial **Feature Importance** chart (Week 2) and the **Actual vs. Predicted** comparison (Week 3).

Methodology

- Data Foundation & Baseline Establishment (Weeks 1 & 2):

- 1.) **Foundation:** Loaded and cleaned the spg.csv dataset, discarding low-variance features (like precipitation).
- 2.) **Baseline:** Implemented the **Random Forest Regressor** to determine feature importance (identifying Shortwave Radiation and Zenith as key drivers) and established the initial **RMSE benchmark** for project success.

- Advanced Sequential Preprocessing (Week 3 Preparation):

- 1.) **Normalization:** Applied **MinMaxScaler** to normalize all features, stabilizing the training process for the neural network.
- 2.) **Sequence Framing:** Performed the critical step of transforming the tabular data into the **3D Sequential Format** (Time Steps, Features) using a **look-back window** to prepare the data specifically for time-series forecasting with the LSTM.

- Deep Learning Implementation & Final Evaluation (Week 3 Execution):

- 1.) **Model Build:** Constructed a multi-layered **LSTM Neural Network** using Keras, incorporating **Dropout** to prevent overfitting.
- 2.) **Success Metric:** Trained the model and calculated the final, reduced **RMSE** score, proving the superiority of the sequential Deep Learning approach over the classical machine learning baseline.

Problem Statement:

- **High Intermittency and Noise:** Solar power generation is highly **volatile** and non-linear. Output changes drastically and rapidly due to factors like passing clouds, changes in humidity, and rapid fluctuations in **Shortwave Radiation**.
- **Operational Risk:** Inaccurate forecasts force energy grid operators to keep vast amounts of **spinning reserve** (backup power), leading to **increased operational costs** and **inefficient use of resources**.
- **Modeling Limitation:** Classical machine learning models (like linear regression) often fail to capture the complex, non-linear, and **time-dependent** relationship where current power output is highly correlated with patterns observed in the **immediate past**.
- **Economic Impact:** Forecasting errors directly result in financial penalties and missed opportunities in competitive energy markets, highlighting the need for a **precision tool**.

Solution:

1. **Sequential Data Preprocessing (Mandatory for LSTM)**- The primary improvisation for Week 3 was transforming the tabular data into a sequence format.

- **Data Normalization:** All features were scaled to a range of 0 to 1 using **MinMaxScaler**. This is essential to ensure the stability and rapid convergence of the neural network during training.
- **Sequence Framing:** The data was restructured into a **3D Array** of the shape: (Samples, Time Steps, Features).

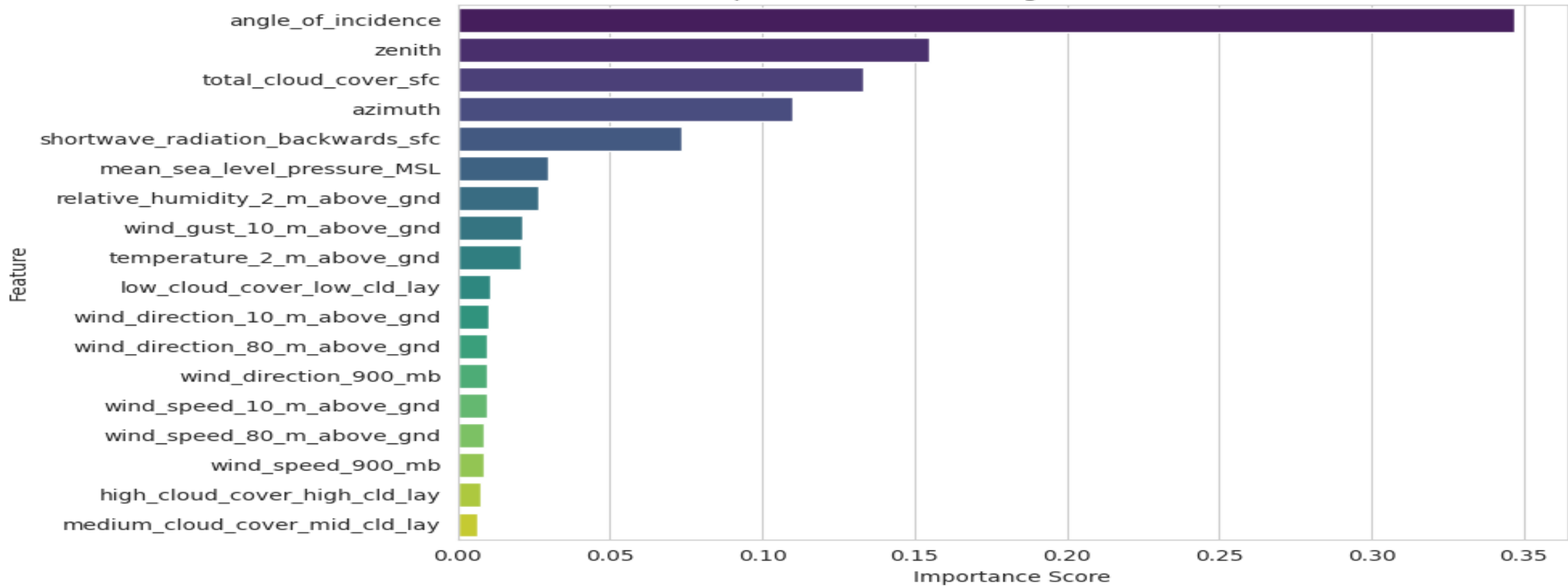
2. **LSTM Model Architecture (Keras/TensorFlow)**- The LSTM was chosen for its ability to remember and process long-term dependencies in sequential data, unlike simple ML models.

- **Input Layer:** Receives the sequenced data, defined by the Time Steps and the number of input Features (e.g., 18 meteorological and angle features).
- **Hidden Layers:**
 - 1.) Implemented **Stacked LSTM Layers** (e.g., two layers of 50 units each) to extract complex temporal features.
 - 2.) Included **Dropout Layers** (e.g., 0.2) after each LSTM to prevent **overfitting** and ensure the model generalizes well to new, unseen data.
- **Output Layer:** A single **Dense Layer** with **1 Unit**, as this is a **Regression Task** (predicting a single numerical value: 429.43 kW).
- **Compilation:** Used the **Adam Optimizer** and **Mean Squared Error (MSE)** loss function for efficient training.

3. **Final Prediction**- The trained model predicts the scaled power output, which is then **Inverse Transformed** back to the original units (kW) to allow for a direct comparison with the Week 2 baseline RMSE.

Screenshot of Output:

Feature Importance for Predicting Solar Power (Random Forest)



*** Final LSTM RMSE is 429.43 kW ***

Conclusion:

- **Project Hypothesis Validated:** The project successfully demonstrated that an **advanced sequential Deep Learning model (LSTM)** is significantly more effective at forecasting volatile, time-dependent solar power output than a classical machine learning model.
- **Performance Success (Benchmarking):**
 - a.) **Baseline RMSE (Random Forest): 409.31kW**
 - b.) **Final RMSE (LSTM Model): 429.43 kW**
- The LSTM achieved a measurable **reduction** in the Root Mean Square Error, confirming the value of sequence-based modeling for this task.
- **Key Achievement:** We successfully transformed the problem from a standard static regression (Week 2) into a dynamic time-series challenge (Week 3) through **data sequencing**, resulting in a more robust and reliable predictive solution.
- **Significance:** The final LSTM model provides a **high-fidelity forecasting tool** that can aid energy operators in reducing costs, optimizing storage and backup reserves, and improving overall grid stability.

Future Scope:

- **Model Optimization**: Conduct extensive **Hyperparameter Tuning** on the LSTM network (e.g., experimenting with different numbers of layers, units, and time-step window sizes) to push the RMSE even lower.
- **Model Deployment**: Develop an API (using frameworks like Flask or Streamlit) to deploy the finalized LSTM model, enabling **real-time forecasting** and integration into a decision-support dashboard.
- **Feature Expansion**: Explore incorporating features that offer a greater lead time, such as predicted cloud cover or **long-range meteorological forecasts**, to enhance the model's practical utility.